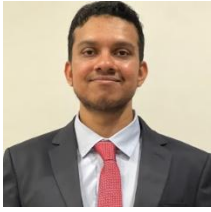


Zoheb Faisal

Zachry Dept. of Civil and Environmental Engineering, 808W DLEB Texas A&M University

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RESEARCH INTERESTS

- 3D Printing Construction
- Durability of Sustainable Building Materials
- Soil Erosion
- Ground Improvement Techniques

EDUCATION

- 2021 – Texas A & M University, TX, USA
Doctor of Philosophy (**Ph.D.**) in Civil Engineering
GPA – 4.0/4.0
Advisor – Anand J. Puppala, Co-Advisor – Petros Sideris
Dissertation – ‘Mechanical and Durability characteristic of 3D-printed building material – Hempcrete’
- 2018 – 2020 College of Engineering Trivandrum, India
Master of Technology (**M.Tech.**) in Geotechnical Engineering
GPA – 8.85/10
Advisor – Jayasree P.K.
Thesis – ‘A Study on Erodibility of Lateritic Slopes due to Flooding’
- 2013 – 2017 College of Engineering Trivandrum, India
Bachelor of Technology (**B.Tech.**) in Geotechnical Engineering
GPA – 8.3/10
Advisor – N. Unnikrishnan
Project – ‘Ground Improvement of Clay using Pre-Stressed Geo-Textiles’

PUBLICATIONS

Journal Papers [In Preparation]

- **Faisal, Z.**, Huang, J., Sideris, P., Puppala, A.J., TBD. Mechanical characterization and non-destructive evaluation of cast and 3D-printed hempcrete for built infrastructure. *(Based on Faisal, Z. completed thesis work for his doctoral degree)*
- **Faisal, Z.**, Huang, J., Sideris, P., Puppala, A.J., TBD. Durability and fire performance of 3D-printable hempcrete under different environmental stressors for built infrastructure. *(Based on Faisal, Z. completed thesis work for his doctoral degree)*
- **Faisal, Z.**, Huang, J., Sideris, P., Puppala, A.J., TBD. Fresh and hardened properties of 3D-printable hempcrete: influence of hemp particle size, hemp dosage, and silica fume on rheology, mechanical strength, and shrinkage. *(Based on Faisal, Z. completed thesis work for his doctoral degree)*

Conference Papers

- **Faisal, Z.**, Muhammed Shibil, P., Jayasree, P.K. and Jayamohan, J., 2021. A Study on the Erodibility of Lateritic Soil Due to Flooding. In Proceedings of the Indian Geotechnical Conference 2019: IGC-2019 Volume I (pp. 649-657). Springer Singapore. DOI: https://doi.org/10.1007/978-981-16-2260-1_32 *(Based on Faisal, Z. completed thesis work for his master's degree)*
- **Faisal, Z.**, Muhammed Shibil, P., Jayasree, P.K. and Jayamohan, J., 2021. Effect of Flood Parameters on Lateritic Slope Erosion. In Transportation, Water and Environmental Geotechnics: Proceedings of Indian Geotechnical

Conference 2020 Volume 4 (pp. 329-335). Springer Singapore. DOI: https://doi.org/10.1007/978-981-33-6346-5_56 (Based on Faisal, Z. completed thesis work for his master's degree)

- Tanshette, A.A., **Faisal, Z.**, Huang, J., Puppala, A.J., 2026. Durability Evaluation of Soil-Geopolymer Based Composites for 3D Printing Applications in Geotechnical Engineering. Geo-Congress 2026 (pp. 409-421). DOI: <https://doi.org/10.1061/9780784486757.040>

Conference Papers [In Preparation]

- Sideris, P., Syed, M., Smith, D., El-Tahlawi, M., Sharma, S., **Faisal, Z.**, Huang, J., Puppala, A.J., Koliou, M. Dixit, M., Yan, W. and Grasley, Z., 2026. Hazard resilient and environmentally sustainable 3D printed structures using hempcrete. 4th International Conference on Natural Hazards and Infrastructure, June 2026, Chania, Greece.
- Smith, D., El-Tahlawi, M., Sharma, S., Syed, M., Sideris, P., **Faisal, Z.**, Koliou, M. and Puppala, A.J., 2026. Out-of-plane testing of a full-scale 3D-printed hempcrete wall. 13th National Conference on Earthquake Engineering 2026, Portland, OR, USA: Earthquake Engineering Research Institute.
- Smith, D., El-Tahlawi, M., Syed, M., Sharma, S., **Faisal, Z.**, Sideris, P., Koliou, M., Puppala, A.J., 2027. Experimental Structural Performance Evaluation of Sustainable, 3D-Printed Hempcrete Walls under Out-of-Plane Loading. ASCE 2027.
- **Faisal, Z.**, Huang, J., Sideris, P., Puppala, A.J., 2027. Mechanical and Durability Characteristics of Hempcrete. ASCE 2027.

Conference Presentations [Presenting Authors Underlined]

- Muhammed Shibil P., Faisal, Z., Jayasree, P. K. and Jayamohan, J., 2019. A review on the effects of flooding on roads above embankments. International conference on Geotechnics for High-Speed Corridors-GHC 2019, Hotel White Dammar, Thiruvananthapuram, India, 24th –26th July 2019.
- Faisal, Z., A.P., Naem, E.K., Rashid, A., Sidharth, V.S., Akshay Kumar, Unnikrishnan, N., Jayamohan, J., 2017. Applications of Pre-Stressed Geotextiles - A state of Art. International Conference on Geo-techniques for Infrastructure Projects GIP-2017, Mascot Hotel, Thiruvananthapuram, India, 27th – 28th February 2017.
- Faisal, Z., Huang, J., Sideris, P., Puppala, A.J., 2025. Mechanical and Durability Characteristics of Carbon-negative 3D-Printable Hempcrete. Engineering Mechanics Institute Conference (EMI 2025), Anaheim, California, USA, 27th – 29th May 2025.

Poster Presentation [Presenting Authors Underlined]

- Faisal, Z., Huang, J., Sideris, P., Puppala, A.J., 2024. Mechanical and Durability Characteristics of 3D Printable Building Material – Hempcrete. Advisory Panel Symposium, Zachry Department of Civil and Environmental Engineering, Center of Infrastructure Renewal, Texas A&M University, September 2024.
- Faisal, Z., Tanshette, A.A., Huang, J., Puppala, A.J., 2025. Mechanical and Durability Characteristics of Soil-Geopolymer Composites (SGC). Infrastructure Advancement Institute (IAI) Summit, 19th – 21st May, 2025, Houston, Texas, USA.
- Faisal, Z., Tanshette, A.A., Huang, J., Puppala, A.J., 2026. Mechanical and Durability Characteristics of Soil-Geopolymer Composites (SGC). Advisory Panel Symposium, Zachry Department of Civil and Environmental Engineering, Center of Infrastructure Renewal, Texas A&M University, February 2026.
- Smith, D., El-Tahlawi, M., Sharma, S., Syed, M., **Faisal, Z.**, Sideris, P., Koliou, M. and Puppala, A.J., 2026. Experimental Testing of Full-Scale 3D Printed Hempcrete Walls for Sustainable Residential Housing – Preliminary Results. Advisory Panel Symposium, Zachry Department of Civil and Environmental Engineering, Center of Infrastructure Renewal, Texas A&M University, February 2026.
- Tanshette, A.A., **Faisal, Z.**, Huang, J., Puppala, A.J., 2026. Durability Evaluation of Soil-Geopolymer Based Composites for 3D Printing Applications in Geotechnical Engineering. Geo-Congress 2026, Salt Lake City, Utah, USA, 9th – 12th March 2026.

Patent

- Patent Application Number: 19/324,787; Inventors: Sideris, P., Grasley, Z., Wang, X., Puppala, A.J., **Faisal, Z.**, Title: “3D-printable or pourable hempcrete composition, resulting building material, and methods of making or using the same”; Type: Utility-Nonprovisional Application under 35 USC 111 (a); Submission Date: September 10, 2025; Review; Current Status: Under Review

Invention Disclosure

- Sideris, P., **Faisal, Z.**, Puppala, A.J. (August 2025). “Pourable Hempcrete for Construction and Beyond”, Submitted via the Texas A&M Innovation

RESEARCH EXPERIENCE

Texas A&M University, College Station, Texas, USA

(June 2023 – Present)

Graduate Assistant Research

Advisor – Anand J. Puppala, Ph.D.

Co-Advisor – Petros Sideris, Ph.D.

- **Hempcrete 3D Printed Buildings for Sustainability and Resilience** (December 2022 – January 2026)
Funded by ARPA-E, DOE
 - Explored the feasibility of various mechanical and durability tests on new material – Hempcrete
 - Developed the testing protocol for Hempcrete
 - Developed a flowable mix of Hempcrete which is used as infill for 3D-printed Hempcrete walls
 - Assisted in constructing a 10-ft 3D-printed wall using a construction-grade 3D-printer

TEACHING EXPERIENCE

Texas A&M University, Texas, USA

(September 2022 – May 2023)

Graduate Assistant Teaching

- **CVEN 365 (Introduction to Geotechnical Engineering)**, Fall 2022, Spring 2023
 - Taught analysis part of geotechnical laboratory experiments
 - Supervised students performing laboratory experiments
 - Graded lab reports

INDUSTRY EXPERIENCE

Logic Soil Survey, Kollam, India

(March 2021 – August 2021)

Geotechnical Engineer

- Write Geotechnical Interpretative Reports
- Calculated Soil Bearing Capacity and settlements for shallow foundations
- Calculated pile capacity of bored in-situ piles and under-reamed piles
- Conducted liquefaction analysis

NestAbide, Thiruvananthapuram, India

(September 2020 – August 2021)

Consultant / Mentor

- Providing mentorship to recruits and acting as a consultant to NestAbide for amphibious house and foundations.
- Researched, analyzed, designed and estimated an amphibious cark park, amphibious green house and an amphibious house.

HONOURS AND AWARDS

2026	Graduate & Professional School Travel Award at Texas A&M University, College Station, Texas
2024	Terracon Endowed Fellowship at Texas A&M University, College Station, Texas
2023	Terracon Endowed Fellowship at Texas A&M University, College Station, Texas
2023	Zachry Department of Civil & Environmental Engineering Excellence Fellowship at Texas A&M University, College Station, Texas
2022	Zachry Department of Civil & Environmental Engineering Excellence Fellowship at Texas A&M University, College Station, Texas
2021	Zachry Department of Civil & Environmental Engineering Excellence Fellowship at Texas A&M University, College Station, Texas

PROFESSIONAL MEMBERSHIPS/AFFILIATIONS

- Geo-Institute (TAMU Student Chapter, 2021- 2024)
 - Treasurer (May 2023 – August 2024)
- American Society of Civil Engineers (TAMU Student Chapter, 2021- 2024)
- Indian Geotechnical Society (College of Engineering Trivandrum Student Chapter, 2018 -2020)
 - President (2019 – 2020)
- Indian Concrete Institute (College of Engineering Trivandrum Student Chapter, 2013 -2017)
 - Treasurer (2015 – 2016)
 - Vice - President (2016 – 2017)

SKILLS

- Languages
 - Can speak in English, Malayalam and Hindi English
 - Can read English, Hindi, Korean, Arabic and Malayalam
- Ability to code using MATLAB
- Proficient in making custom meal plans and workout plans
- Familiar with software like Plaxis and GeoStudio

REFERNECES

- **Anand J. Puppala, Ph.D.**
 - Professor, Zachary Department of Civil Environmental and Construction Engineering, Texas A&M University, College Station, TX
 - 979-458-6346, anandp@tamu.edu
- **Petros Sideris, Ph.D.**
 - Associate Professor, Zachary Department of Civil Environmental and Construction Engineering, Texas A&M University, College Station, TX
 - 979-845-2708, petros.sideris@tamu.edu
- **Maria Koliou, Ph.D.**
 - Associate Professor, Zachary Department of Civil Environmental and Construction Engineering
 - Texas A&M University, College Station, TX
 - 979-845-4469, maria.koliou@tamu.edu
- **Nripojyoti Biswas, Ph.D.**
 - Assistant Professor, Department of Civil and Environmental Engineering
 - University of Massachusetts Lowell, Lowell, MA

- 978-934-2761, nripojyoti_biswas@uml.edu

- **Jayasree P.K., Ph.D.**

- Professor, Department of Civil Engineering
- College of Engineering Trivandrum, India.
- +91-94472-05324, jayasreepk@cet.ac.in